

TRANSPORTATION SERVICES



OPTIMIZING MARKETPLACE EFFICIENCY
SOLVING IGNORES & CANCELLATIONS

CREDITE: PHALLA SONG

POWER BI DASHBOARD

9.894K

Total Booking

84 %

Completion Rate

16 %

Cancellation Rate

5

Avg Ignored Drivers

3.61

Avg Distance (KM)

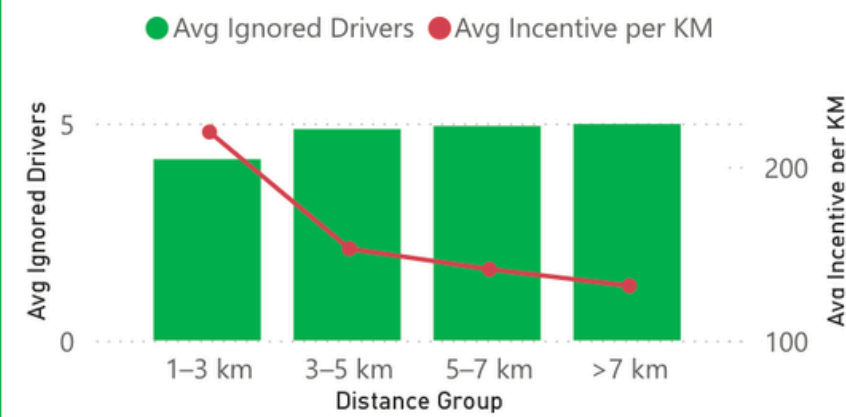
182

Avg Incentive per KM(₹)

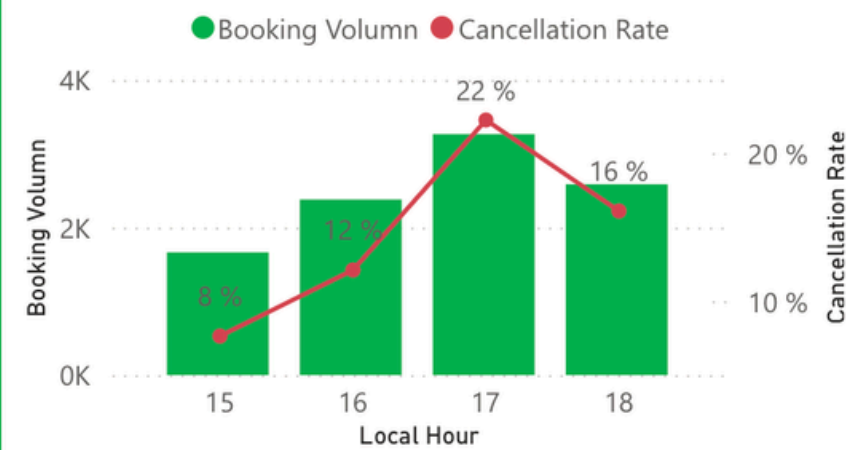
2.05

Avg ETA...

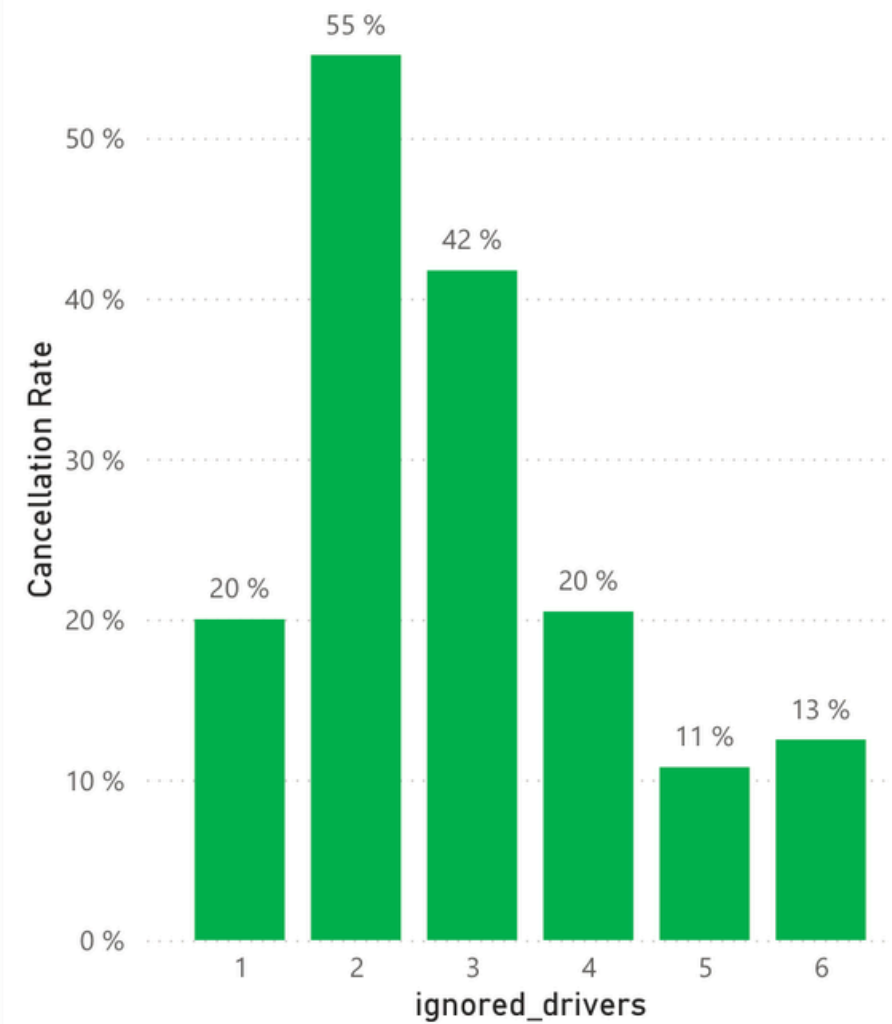
Pricing Problem: Ignored Drivers Vs Intensive/Km



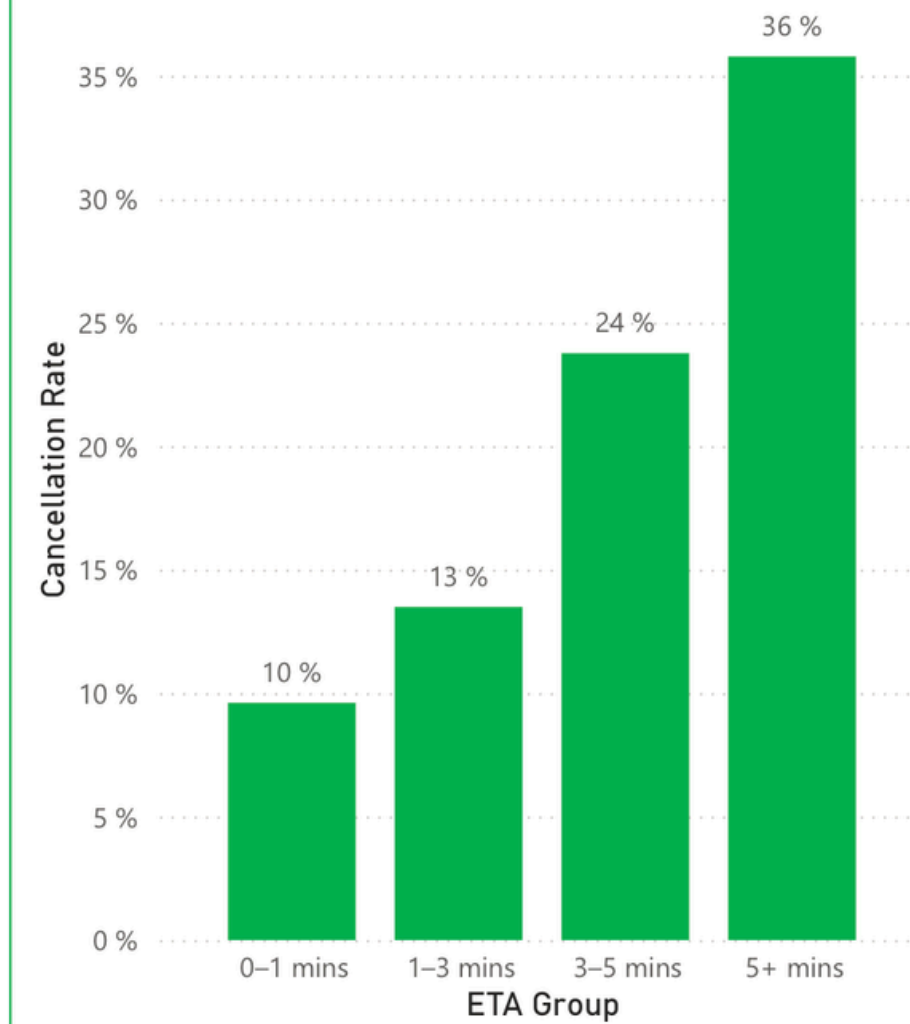
The peak hour problem



Ignored Drivers impact on cancellation rate



ETA effected on cancellation rate



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1. PROBLEM STATEMENT

A significant portion of customer booking attempts fails due to driver rejection and passenger cancellations.

These failures result in:

- Lost revenue
- Lower operational efficiency
- Poorer customer experience
- Reduced platform reliability

Understanding the underlying drivers of failed bookings is critical for improving overall marketplace health and increasing successful trip conversions.

Why this happened?

1. Driver Behavior: What factors cause drivers to repeatedly Ignore a booking?
2. Passenger Behavior: What factors lead to a high probability of a booking being Cancelled?



2. DATA UNDERSTANDING

Dataset Overview

The dataset consists of ride booking attempts with information covering:

- Pickup and dropoff locations
- Trip characteristics (distance, duration, ETA)
- Driver behavior (ignored drivers)
- Booking outcomes (completed vs. canceled)

What This Data Allows Us To Analyze?

- Customer demand behavior (by hour and area)
- Supply availability & friction (ETA, ignored drivers)
- Driver rejection patterns
- Trip profitability variables (distance, incentives)
- Factors influencing booking failure

9,894

TOTAL BOOKINGS

84%

COMPLETION RATE

16%

CANCELLATION RATE

Data Purpose:

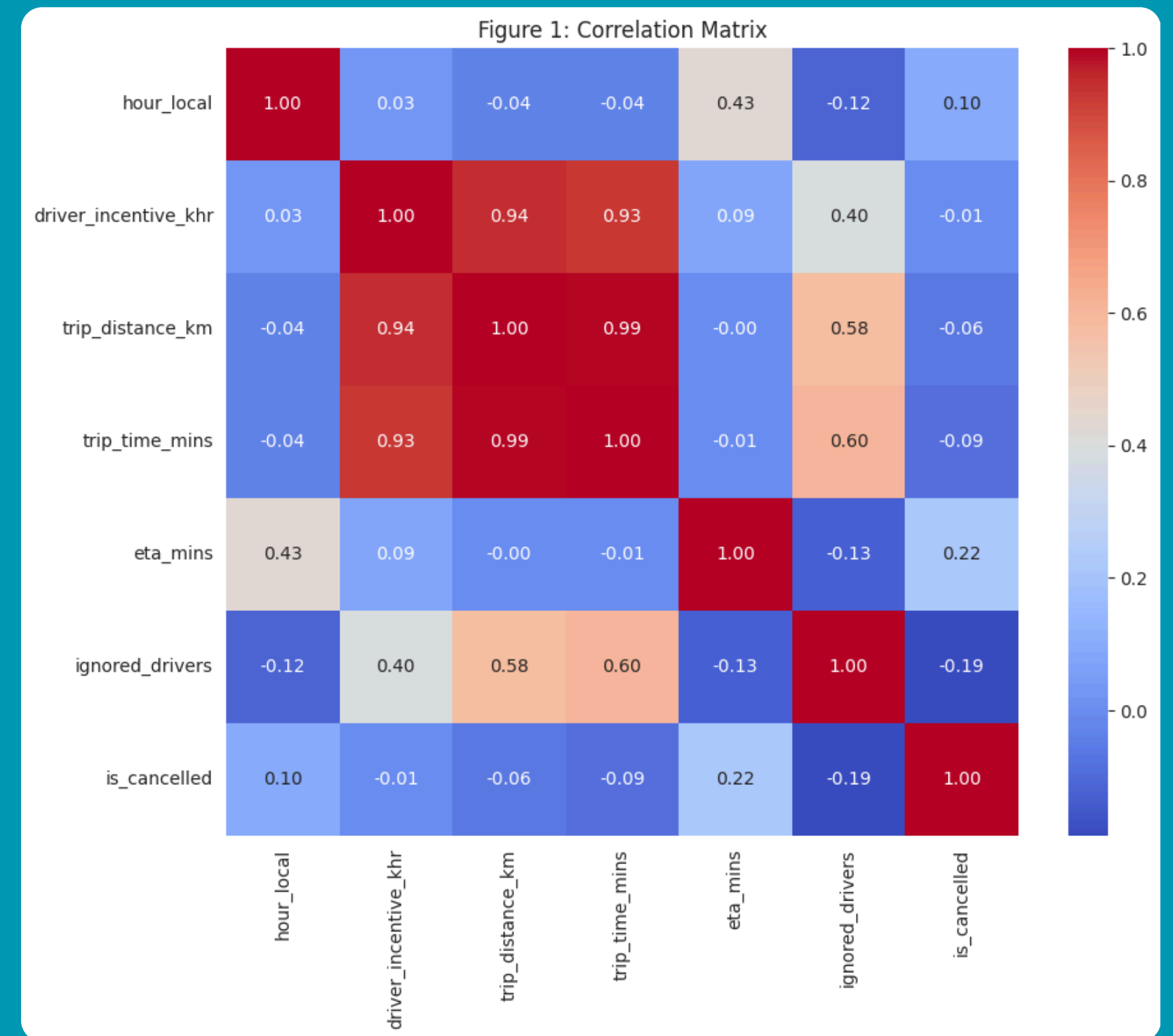
To understand the drivers of booking failure and identify actionable levers to reduce cancellations and improve overall marketplace efficiency.

3. EXPLORATORY DATA ANALYSIS

What Predicts Driver Ignores?

Correlation Analysis:

- **Strongest Correlation: Trip Distance (+ 0.58)**
- **Secondary Correlation: Driver Incentive (+ 0.40)**

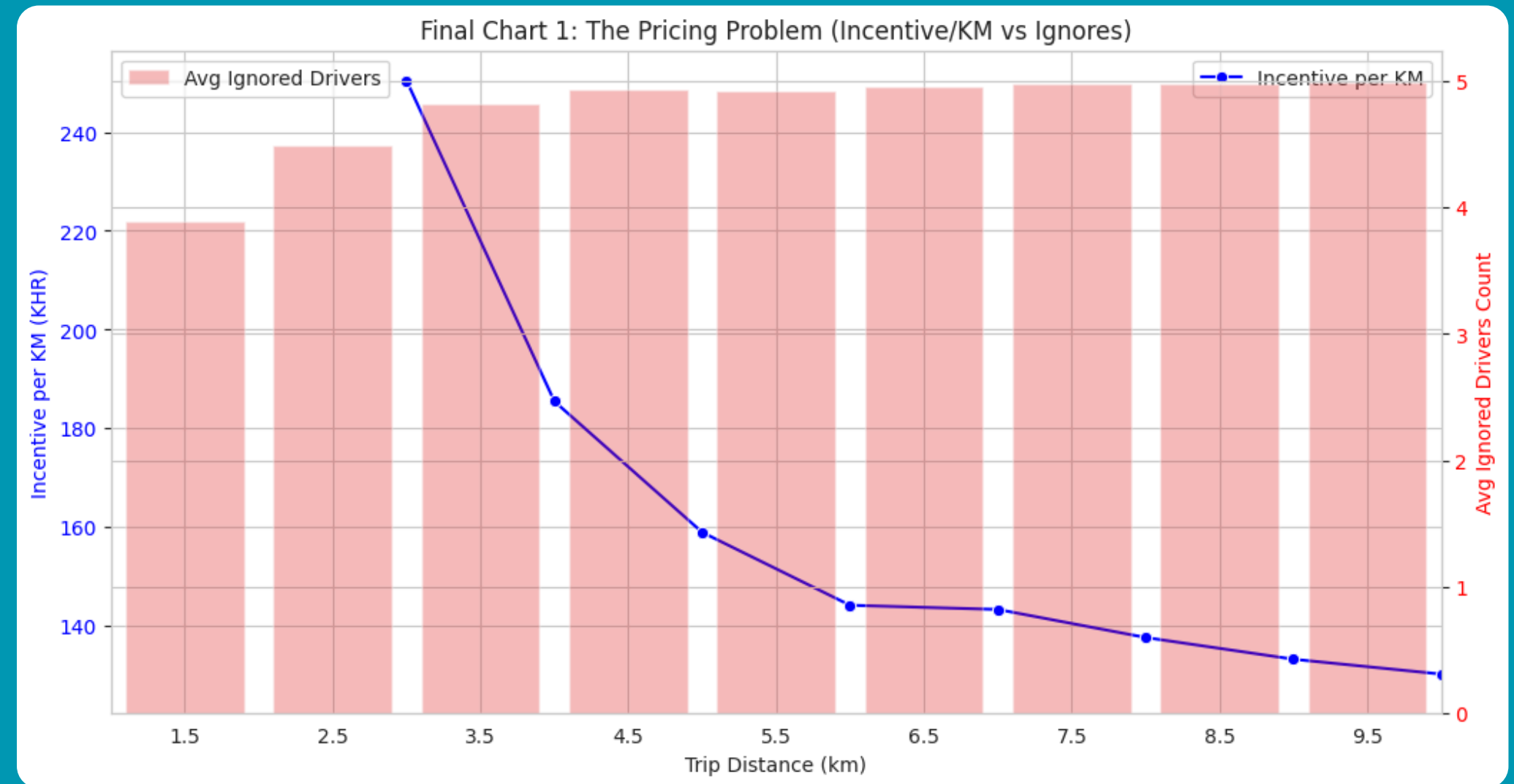


Let's go through a deep analysis to see why it happened in the next slide.

4. KEY FINDINGS

1. Why do drivers ignore booking?

- Observation: Bookings with longer distances are ignored significantly more often.
- Root Cause: The pricing curve is inefficient.
 - Short Trip: 255 KHR per KM.
 - Long Trip: 128 KHR per KM.
- Takeaway: Drivers are optimizing for yield (per km), not just total fare.

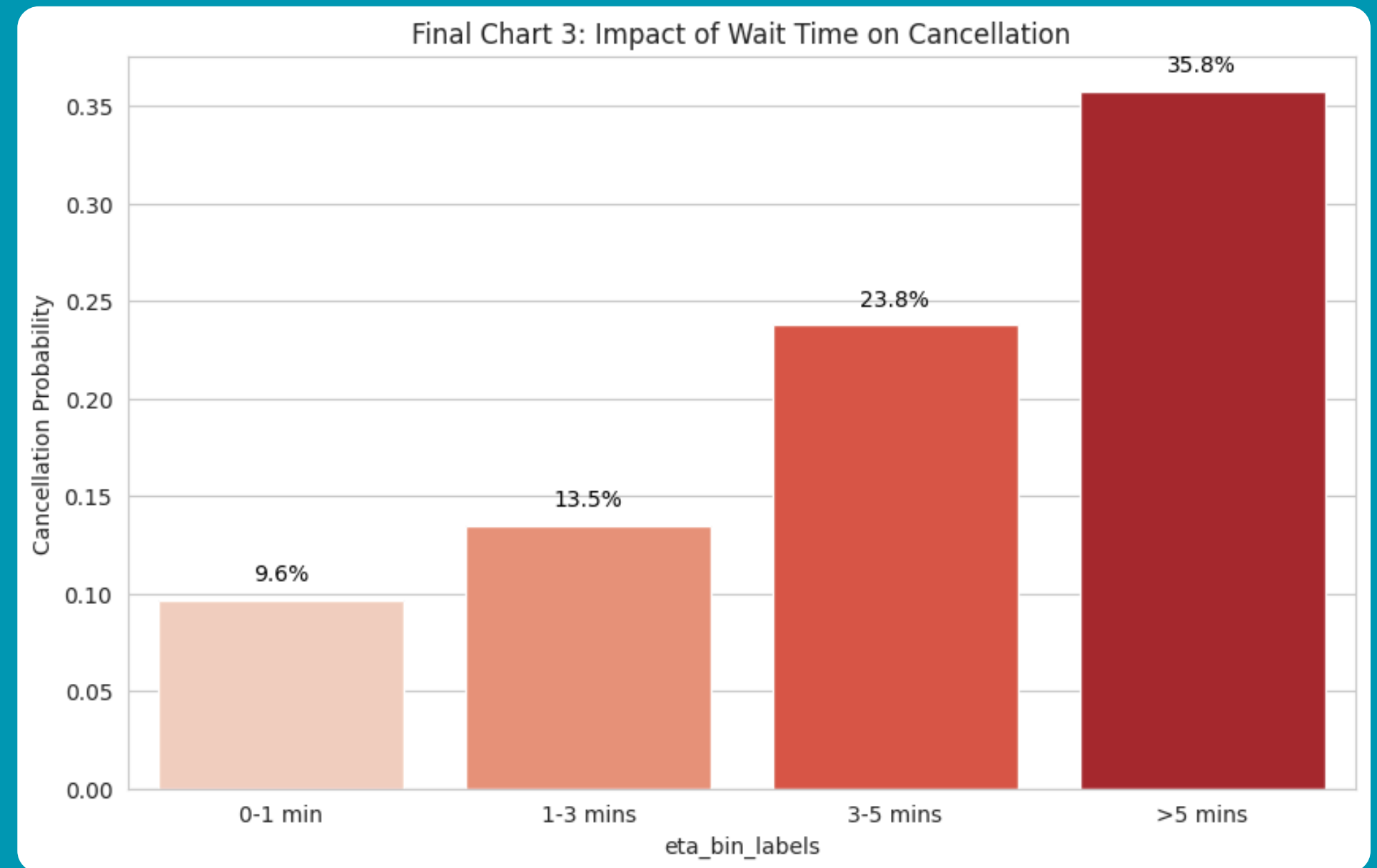


As the earnings per kilometer (blue line) drop, the number of ignored drivers (red bars) rises

4. KEY FINDINGS

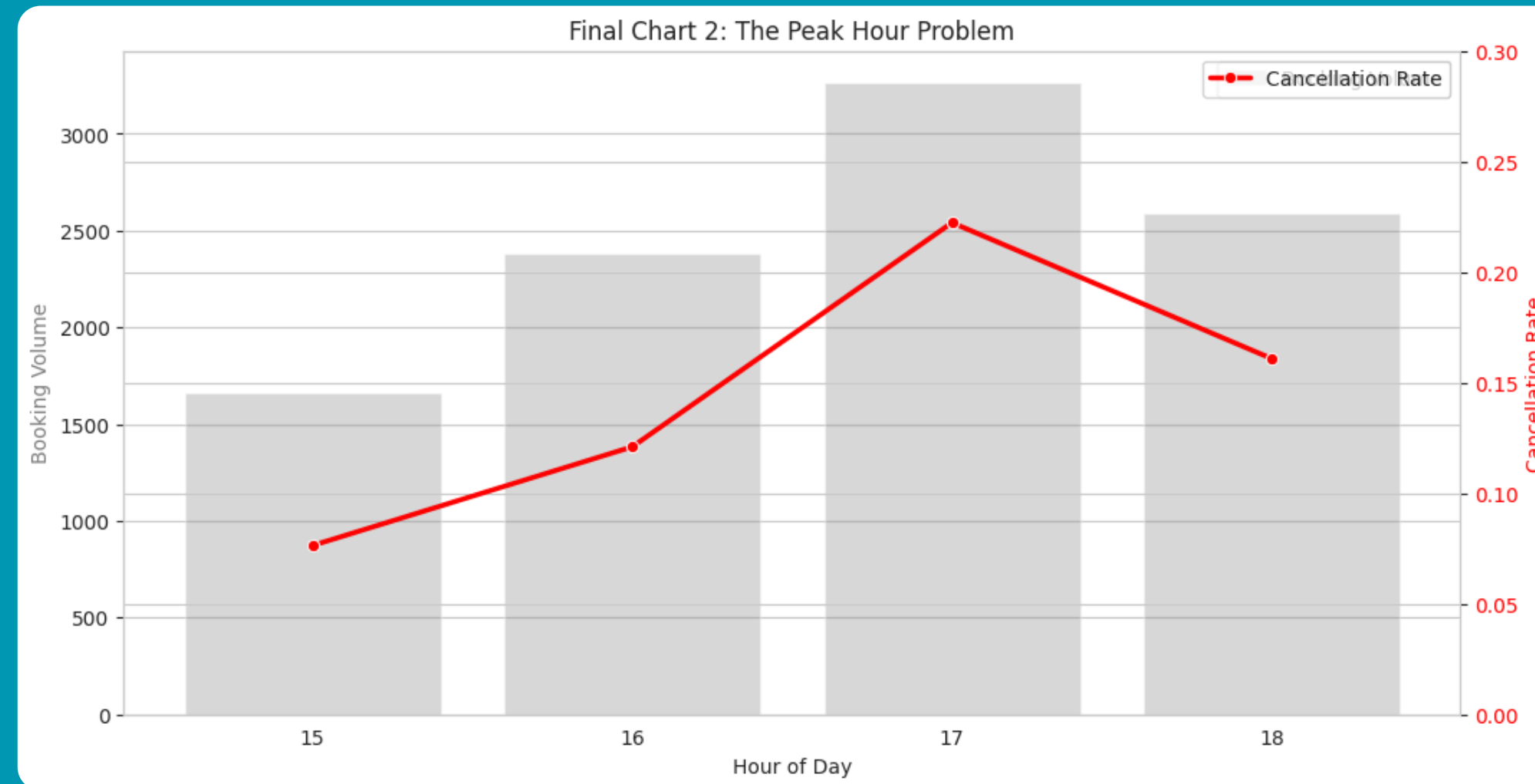
2. Why do Passengers Cancel?

- **Factor: Estimated Time of Arrival (ETA).**
- **Data:**
 - Wait < 1 min: 9.6% Cancel.
 - Wait > 5 mins: 35.8% Cancel.



4. KEY FINDINGS

2. Why do Passengers Cancel?



- The "5 PM Bottleneck"
- Context: 17:00 is the peak hour for demand (3,265 bookings).
- The Crash: High demand leads to scarcity -> High Cancellation (22%).

5. STRATEGIC RECOMMENDATIONS

Flatten the Curve:

Keep Incentive per KM above 200 KHR for long trips to reduce ignores on high-distance routes.

Own the Peak

Launch a “5 PM Power Hour” operational playbook to temporarily increase driver supply.

Smart Matching Logic

Enhance the dispatch algorithm to expand search radius faster when the first driver ignores the request.

Implementation

- Introduce a Long-Trip Boost: +20–30% incentive for >5 km rides

- Targeted 1-hour surge boosts in high-demand hotspots
- Push reminders to bring idle drivers online
- Stackable micro-bonuses for 2–3 quick consecutive trips
- Driver heatmap highlighting high-bonus zones

- +20% radius after 1 ignore
- +40% radius after 2 ignores
- Auto-expand radius for high-ETA bookings (>5 mins)
- Route to drivers with strong recent acceptance

Expected impact

- Reduce ignored drivers by 15–25% in long-distance zones
- Improve ETAs and customer wait time

- Increase driver supply by 10–15% during peak
- Reduce cancellations by 20–30%
- Shorten ETA by 1–2 minutes, improving customer experience

- Reduce ignored-driver count by 1–1.5 on average
- Reduce cancellations by 10–15%
- Improve overall matching speed

6. CONCLUSION

Key Findings

- **Pricing Inefficiency:** Long trips suffer from a 50% drop in incentive-per-km, causing drivers to ignore them.
- A surge at 17:00 (5 PM) leads to driver scarcity, tripling wait times and spiking cancellations.
- **The Marketplace Danger Zone:** Cancellation probability spikes to 55% after a booking has been ignored by just 2 drivers

Recommendations

- **Dynamic Pricing:** Restructure incentives to maintain > 200 KHR/km for long-distance trips
- **Peak Supply Boost:** Implement a targeted "Power Hour" multiplier for the 17:00-18:00 window.
- **Prioritize speed.** Expand the driver search radius immediately after the first ignore to avoid the "2-ignore" cliff.

Better Pricing. Better Supply. Happy Customers.